# Smart Intrusion Detection & Threat Monitoring System (SIDTMS)

A report Submitted in partial fulfillment of the requirements for the award of degree of

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Computer Science Engineering

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SUBMITTED BY

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Registration number – 12318018

Signature of the Student :

## To whom so ever it may concern

I, Ankit Kumar ,12318018 , hereby declare that the work done by me on “ Smart Intrusion Detection & Threat Monitoring System (SIDTMS)” from February 2026 to March 2026, is a record of original work for the partial fulfillment of the requirements for the award of the degree.

B. TECH CSE

Ankit Kumar (12318018)

Signature of the student:
Dated: 23/03/2026

## Abstract

This paper presents a lightweight intrusion detection and threat monitoring system designed for real-time network analysis. The system captures packets using Python-based tools and applies rule-based detection to identify suspicious activities such as port scanning and abnormal traffic patterns. Detected threats are logged and visualized through a web-based dashboard. The proposed system provides a low-cost alternative to traditional Security Information and Event Management solutions.

Keywords— Intrusion Detection System, Cybersecurity, Network Monitoring, Scapy, Flask

## I. Introduction

With the increasing dependence on digital systems, cyber threats have become a major concern. Intrusion Detection Systems are essential tools that help in identifying malicious activities within a network. This project focuses on developing a simple and efficient IDS using Python technologies to monitor network traffic and detect anomalies.

## II. Methodology

### 1. Packet capturing using Scapy

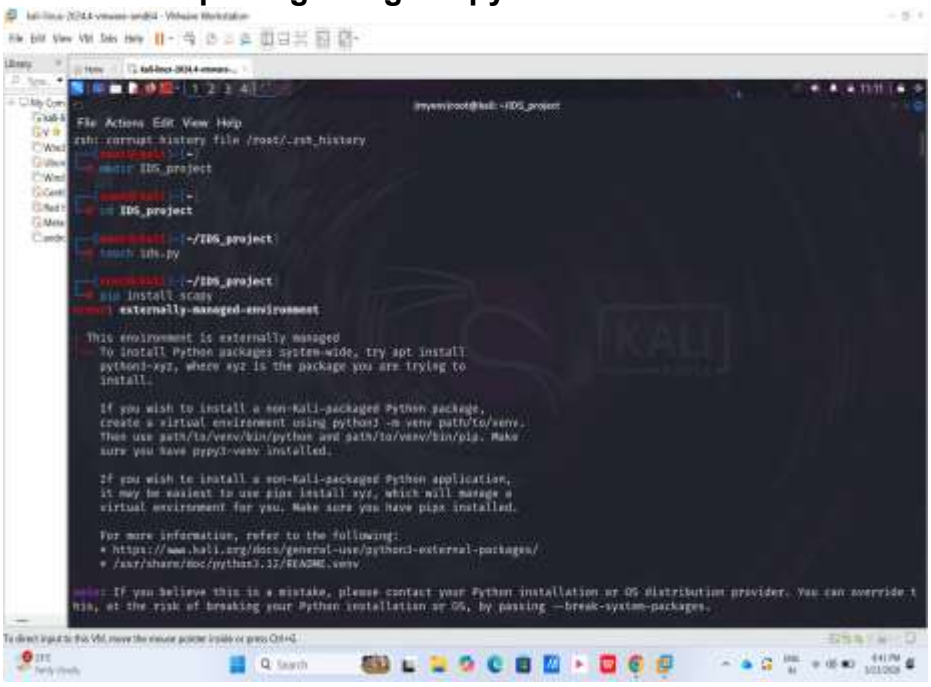



Fig.1: Project Setup in Kali Linux

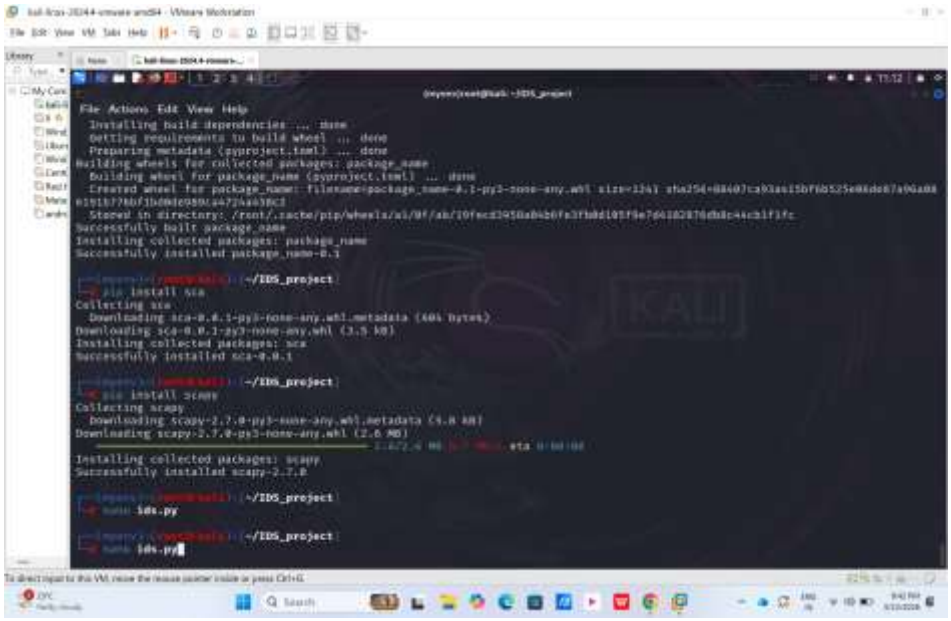



Fig.2: Installing Scapy

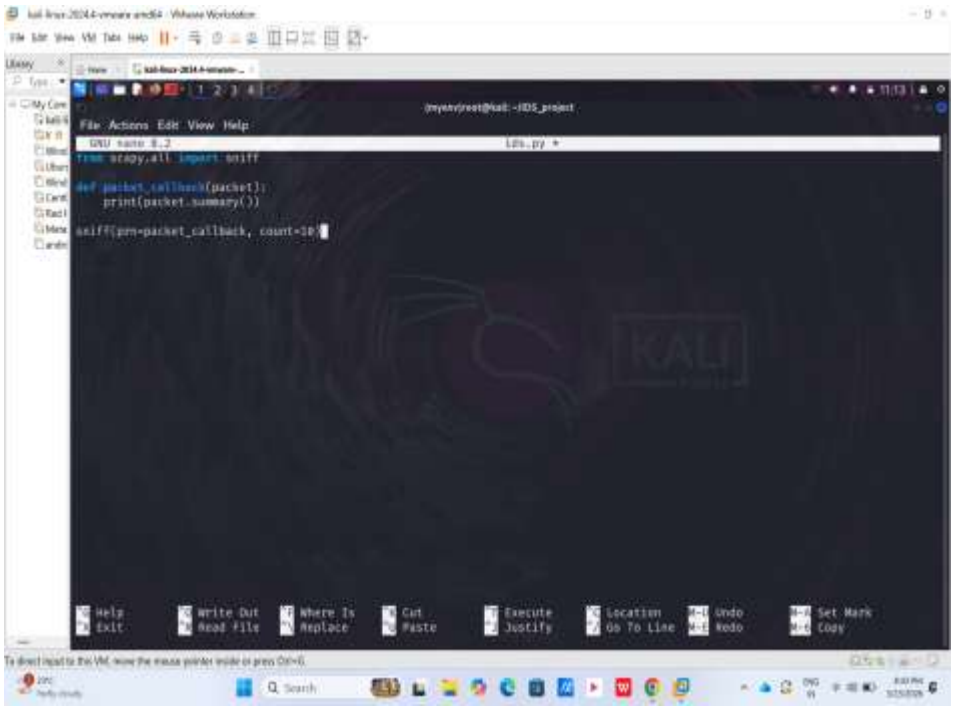



Fig.3: IDS Code Implementation

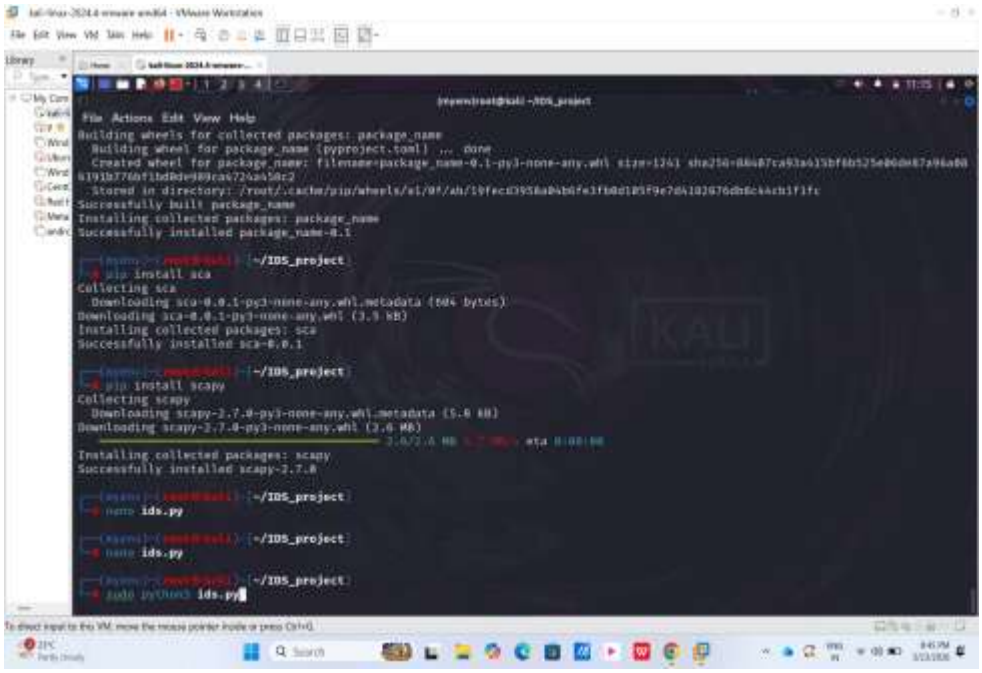



Fig.4: Running IDS Script

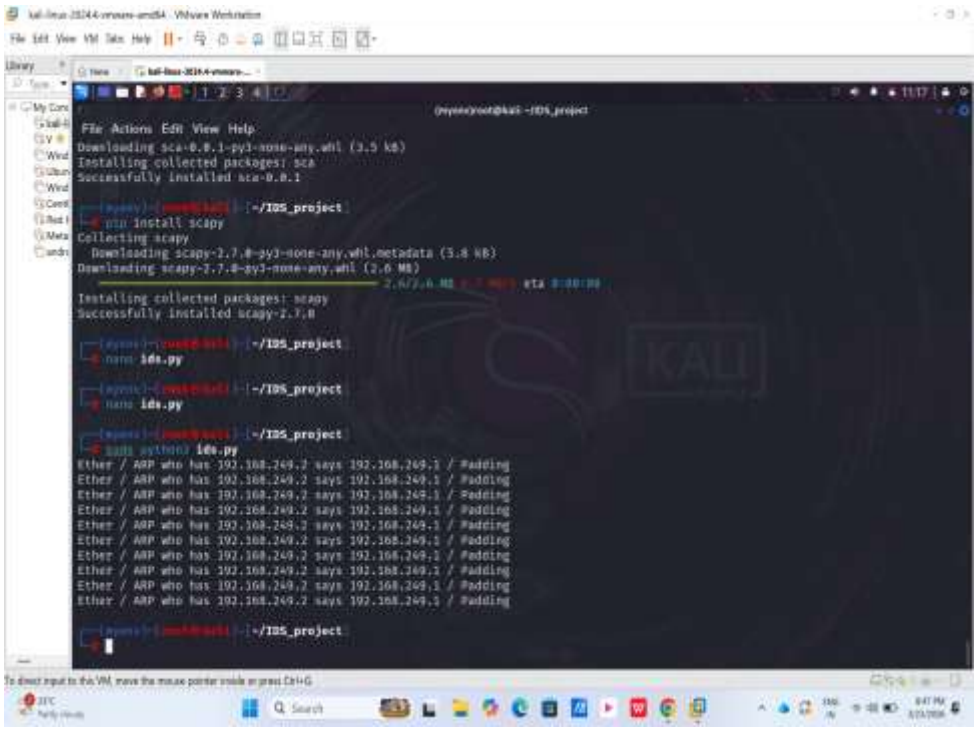
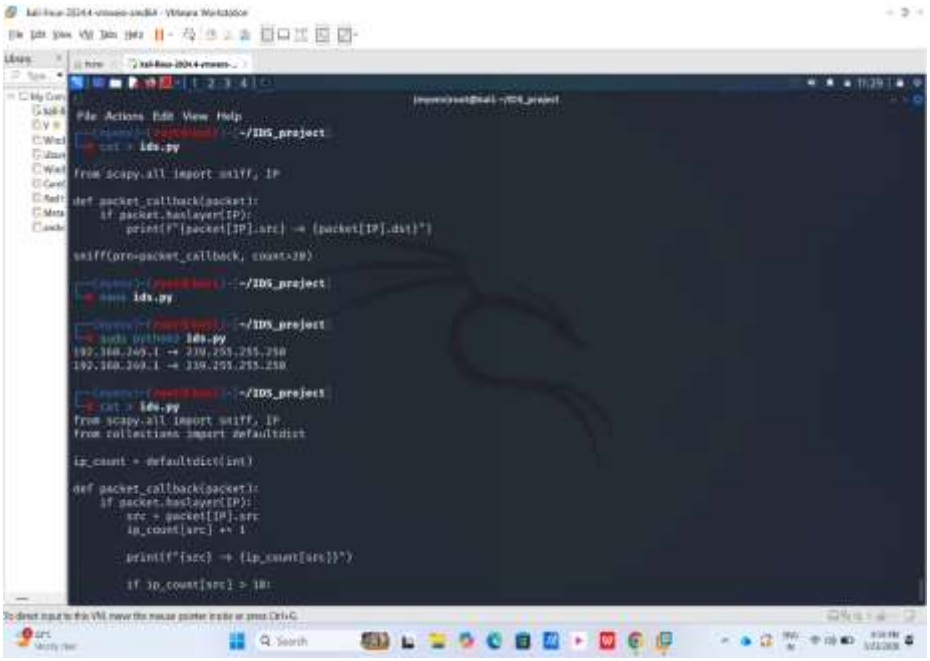
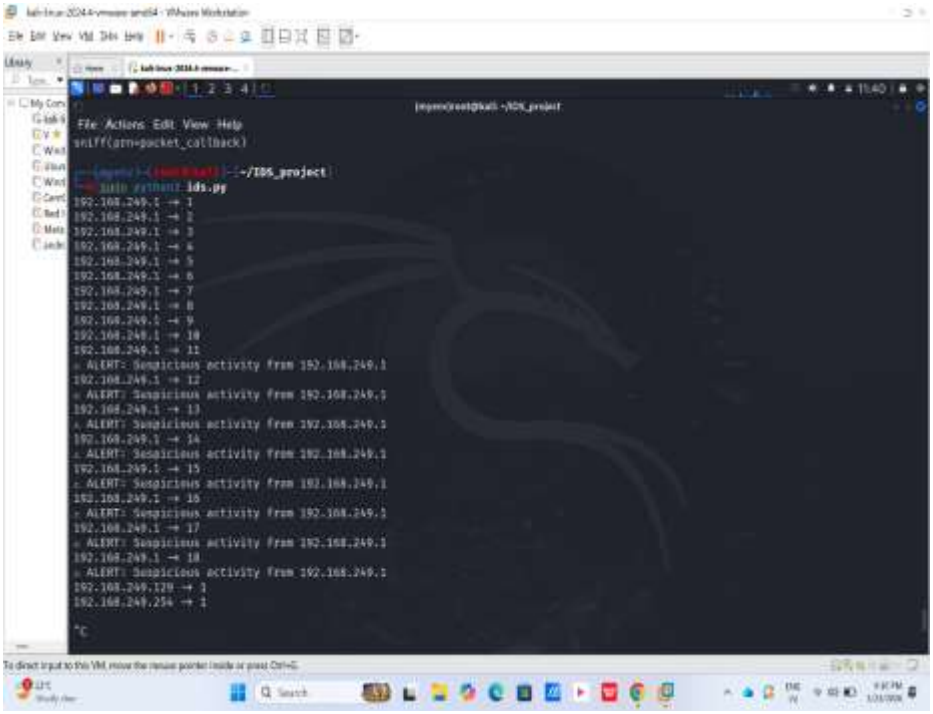



Fig.5: Packet Capture Output

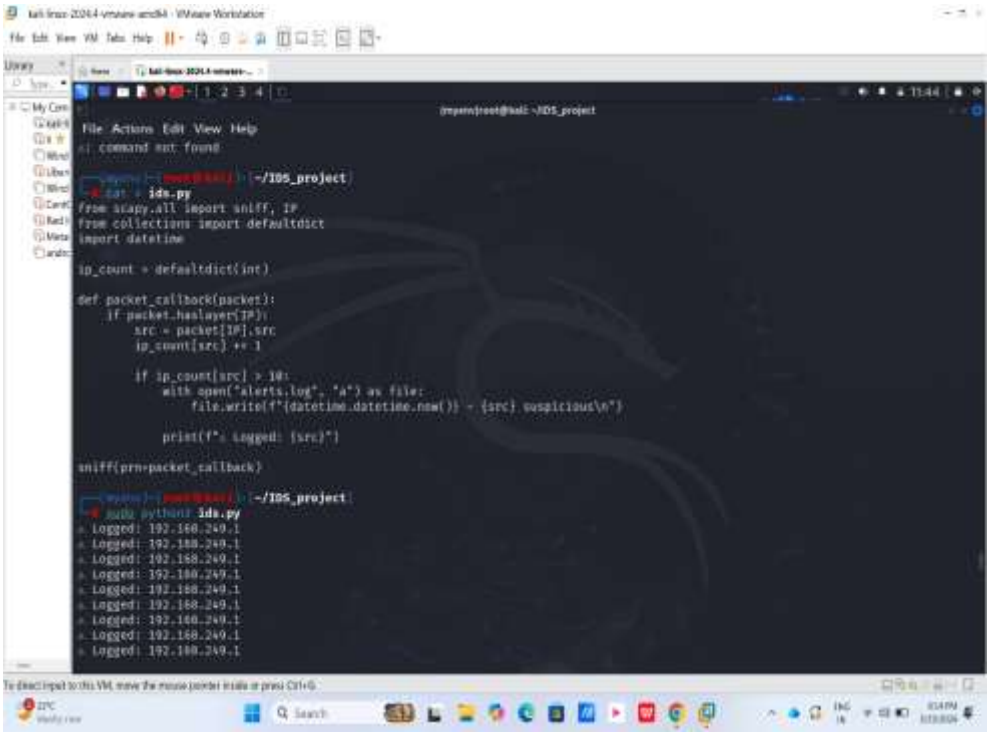
## 2. Extraction of IP and TCP information

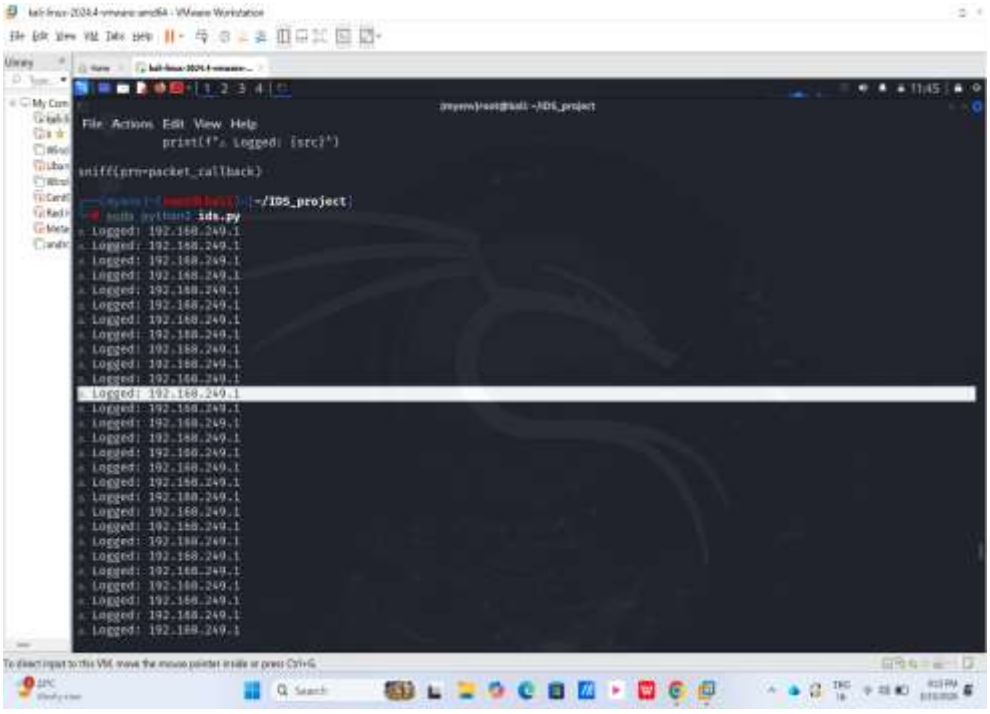


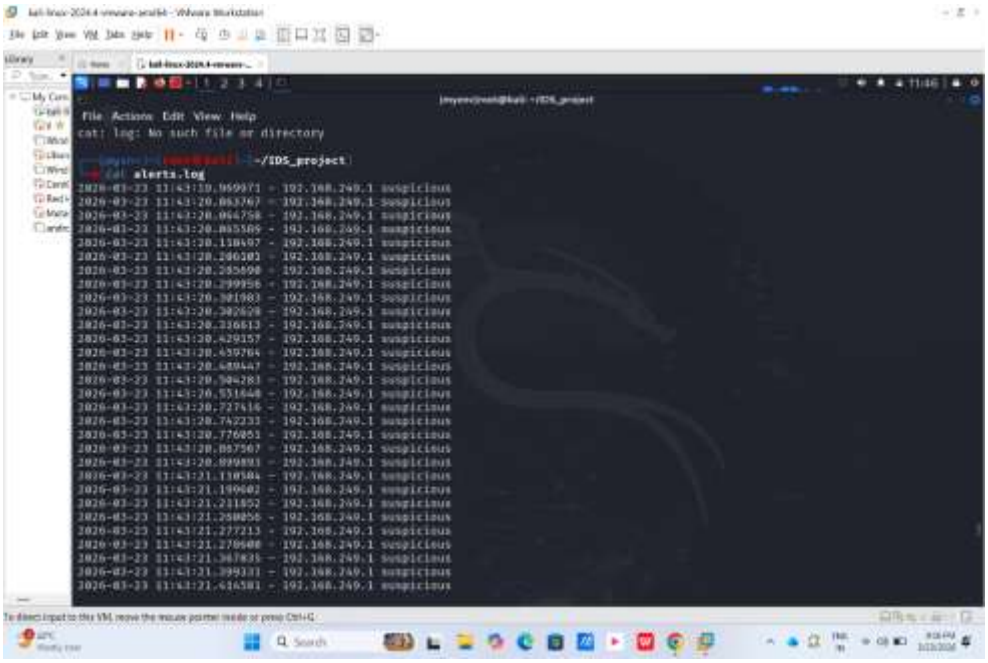

## 3. Application of rule-based detection logic

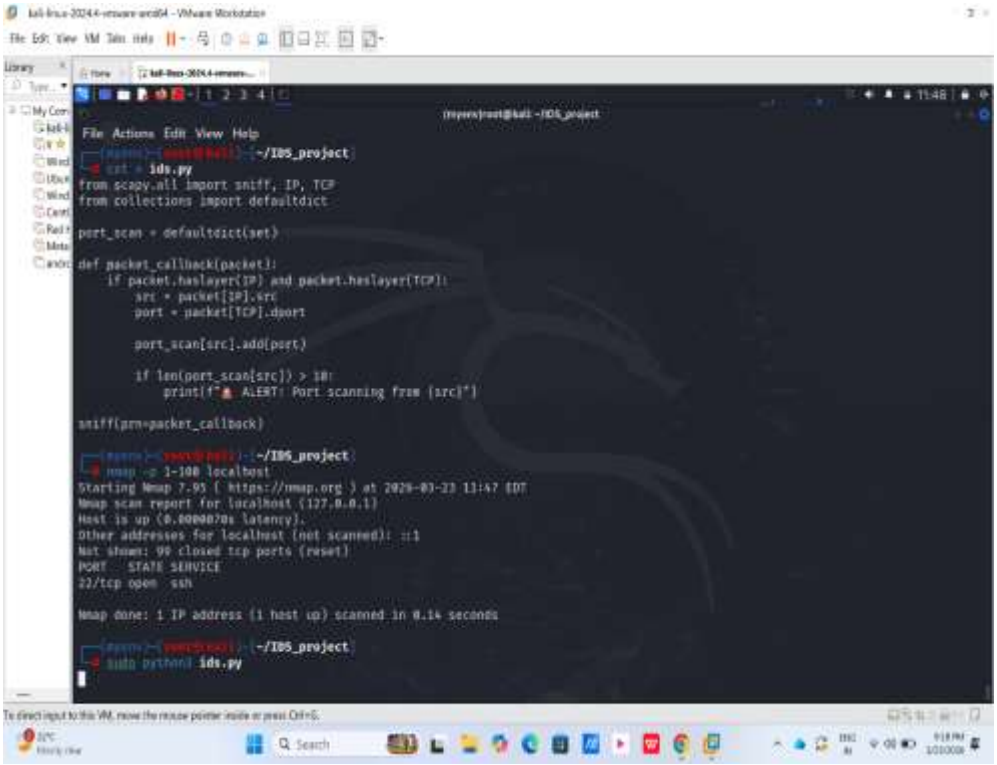


## 4. Logging of suspicious activities

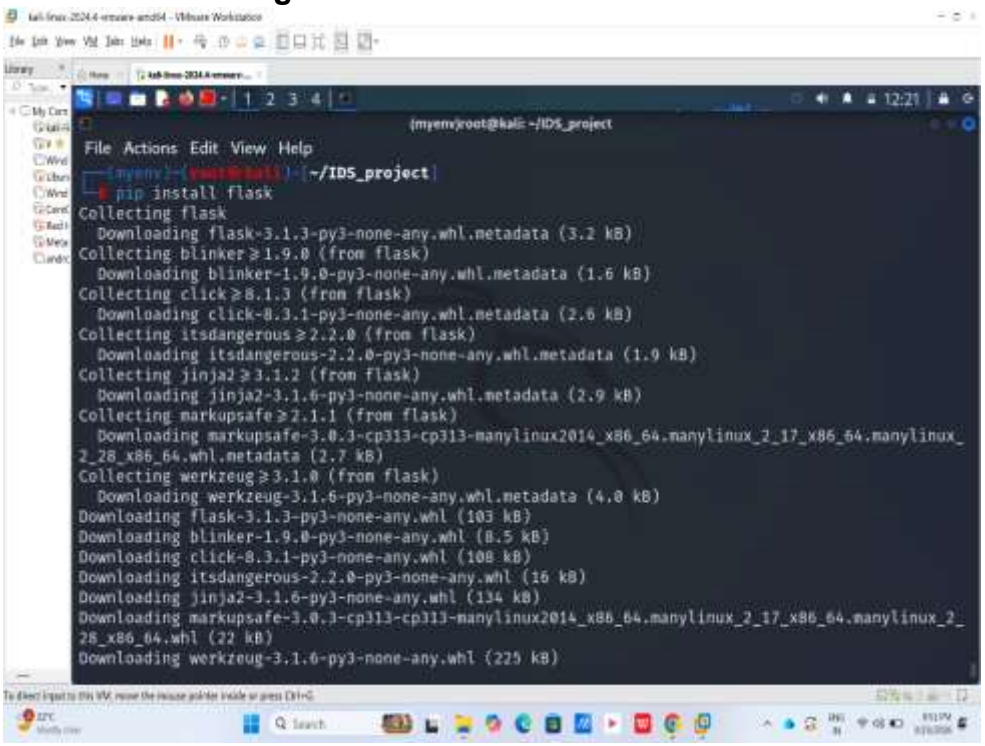


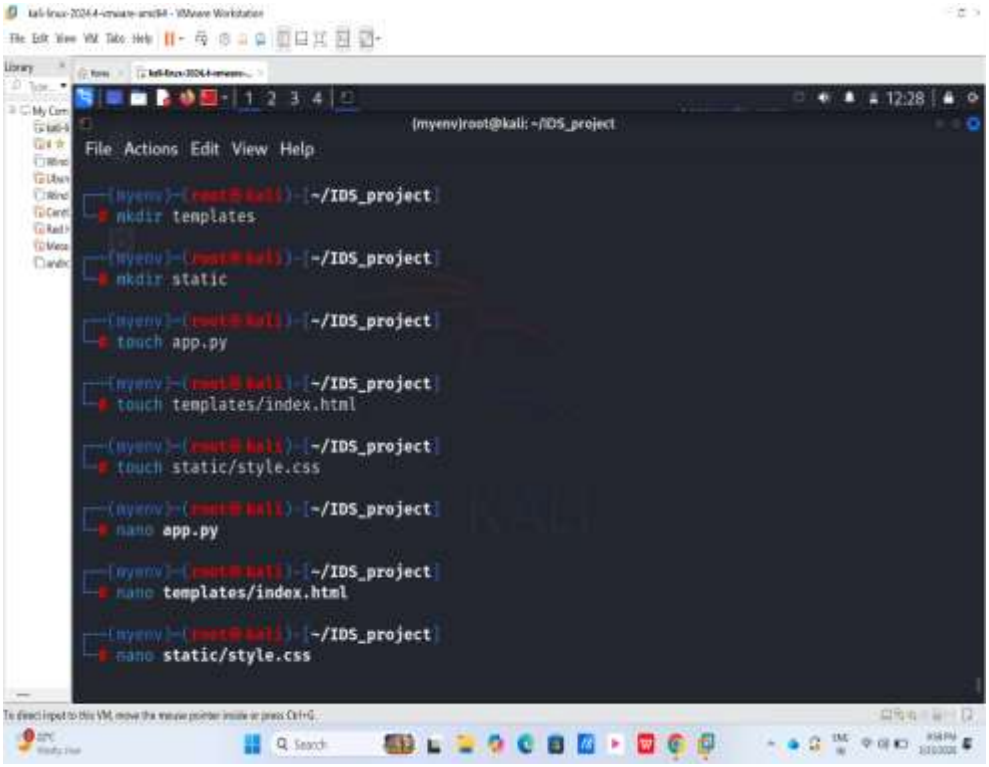



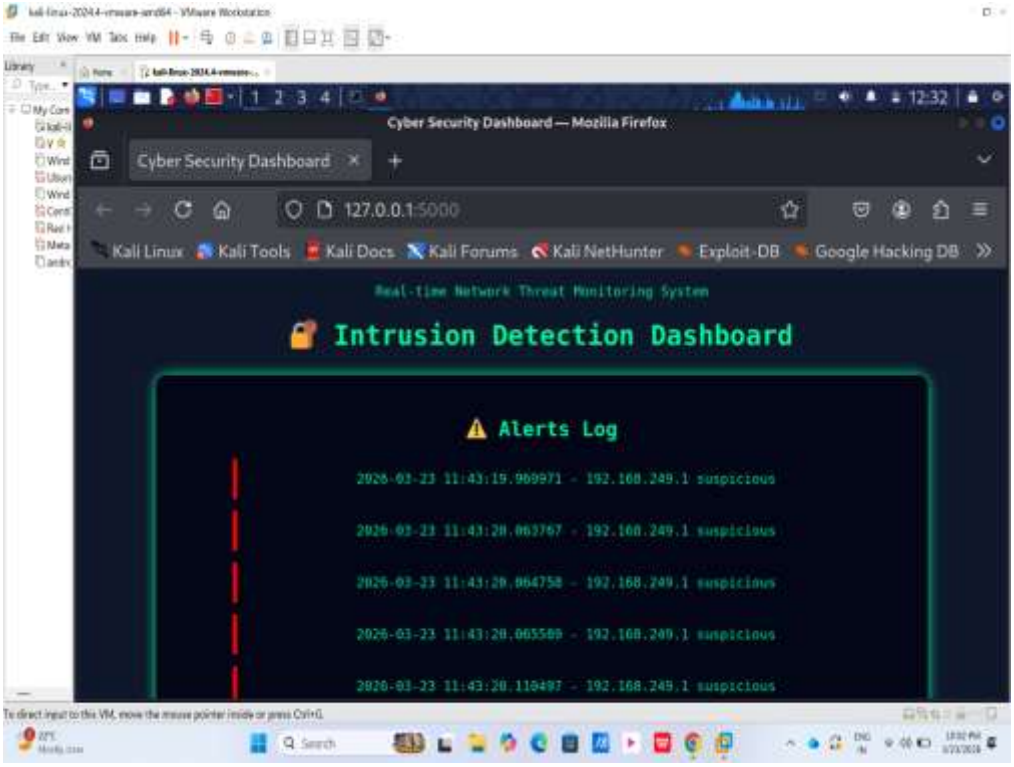

File Actions Edit View Help
cat: log: No such file or directory
~/IDS_project
cat alerts.log
2026-03-23 11:43:19.969971 - 192.168.249.1 suspicious
2026-03-23 11:43:20.063767 - 192.168.249.1 suspicious
2026-03-23 11:43:20.064758 - 192.168.249.1 suspicious
2026-03-23 11:43:20.065589 - 192.168.249.1 suspicious
2026-03-23 11:43:20.118497 - 192.168.249.1 suspicious
2026-03-23 11:43:20.286101 - 192.168.249.1 suspicious
2026-03-23 11:43:20.285698 - 192.168.249.1 suspicious
2026-03-23 11:43:20.299956 - 192.168.249.1 suspicious
2026-03-23 11:43:20.301983 - 192.168.249.1 suspicious
2026-03-23 11:43:20.302628 - 192.168.249.1 suspicious
2026-03-23 11:43:20.336613 - 192.168.249.1 suspicious
2026-03-23 11:43:20.429157 - 192.168.249.1 suspicious
2026-03-23 11:43:20.459704 - 192.168.249.1 suspicious
2026-03-23 11:43:20.489447 - 192.168.249.1 suspicious
2026-03-23 11:43:20.504283 - 192.168.249.1 suspicious
2026-03-23 11:43:20.551648 - 192.168.249.1 suspicious
2026-03-23 11:43:20.727416 - 192.168.249.1 suspicious
2026-03-23 11:43:20.742233 - 192.168.249.1 suspicious
2026-03-23 11:43:20.776051 - 192.168.249.1 suspicious
2026-03-23 11:43:20.867567 - 192.168.249.1 suspicious
2026-03-23 11:43:20.899891 - 192.168.249.1 suspicious
2026-03-23 11:43:21.110584 - 192.168.249.1 suspicious
2026-03-23 11:43:21.199602 - 192.168.249.1 suspicious
2026-03-23 11:43:21.231852 - 192.168.249.1 suspicious
2026-03-23 11:43:21.260856 - 192.168.249.1 suspicious
2026-03-23 11:43:21.277213 - 192.168.249.1 suspicious
2026-03-23 11:43:21.278600 - 192.168.249.1 suspicious
2026-03-23 11:43:21.367016 - 192.168.249.1 suspicious
2026-03-23 11:43:21.399333 - 192.168.249.1 suspicious
2026-03-23 11:43:21.416581 - 192.168.249.1 suspicious

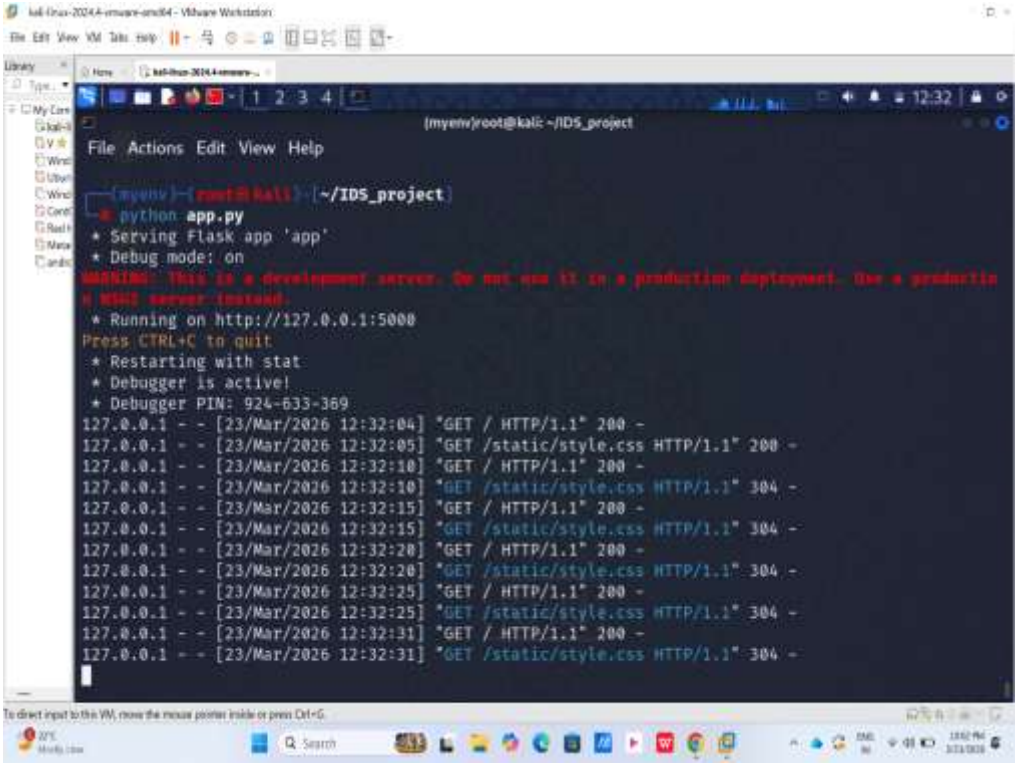

File Actions Edit View Help
~/IDS_project
cat > ids.py
from scapy.all import sniff, IP, TCP
from collections import defaultdict
port_scan = defaultdict(set)
def packet_callback(packet):
if packet.haslayer(IP) and packet.haslayer(TCP):
src = packet[IP].src
port = packet[TCP].dport
port_scan[src].add(port)
if len(port_scan[src]) > 10:
print(f"🚨 ALERT: Port scanning from {src}")
sniff(prn=packet_callback)
~/IDS_project
nmap -p 1-100 localhost
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-23 11:47 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000070s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 99 closed tcp ports (reset)
PORT STATE SERVICE
22/tcp open ssh
Nmap done: 1 IP address (1 host up) scanned in 0.14 seconds
~/IDS_project
sudo python3 ids.py

## 5. Visualization using Flask dashboard




Cyber Security Dashboard — Mozilla Firefox
Cyber Security Dashboard
127.0.0.1:5000
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
Real-time Network Threat Monitoring System
Intrusion Detection Dashboard
Alerts Log
2026-03-23 11:43:19.909971 - 192.168.249.1 suspicious
2026-03-23 11:43:20.063767 - 192.168.249.1 suspicious
2026-03-23 11:43:20.064758 - 192.168.249.1 suspicious
2026-03-23 11:43:20.065589 - 192.168.249.1 suspicious
2026-03-23 11:43:20.110497 - 192.168.249.1 suspicious


(myenv)root@kali: ~/IDS_project
File Actions Edit View Help
(myenv)-(root㉿kali)-[~/IDS_project]
# python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a productio
n WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 924-633-369
127.0.0.1 - - [23/Mar/2026 12:32:04] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [23/Mar/2026 12:32:05] "GET /static/style.css HTTP/1.1" 200 -
127.0.0.1 - - [23/Mar/2026 12:32:10] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [23/Mar/2026 12:32:10] "GET /static/style.css HTTP/1.1" 304 -
127.0.0.1 - - [23/Mar/2026 12:32:15] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [23/Mar/2026 12:32:15] "GET /static/style.css HTTP/1.1" 304 -
127.0.0.1 - - [23/Mar/2026 12:32:20] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [23/Mar/2026 12:32:20] "GET /static/style.css HTTP/1.1" 304 -
127.0.0.1 - - [23/Mar/2026 12:32:25] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [23/Mar/2026 12:32:25] "GET /static/style.css HTTP/1.1" 304 -
127.0.0.1 - - [23/Mar/2026 12:32:31] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [23/Mar/2026 12:32:31] "GET /static/style.css HTTP/1.1" 304 -

## III. System Architecture

The system consists of four major components: packet sniffer, detection engine, logging system, and web dashboard. Network traffic is captured and analyzed to detect threats, which are then logged and displayed on the dashboard.

## IV. Results

The system successfully captures network packets and detects suspicious activities such as port scanning. Logs are generated and displayed on the dashboard, demonstrating real-time monitoring capabilities.

## V. Conclusion

The developed IDS provides an effective and low-cost solution for monitoring network security. It demonstrates how basic cybersecurity mechanisms can be implemented using Python. Future enhancements may include machine learning-based detection and cloud integration.

## References


1 Python Software Foundation, Python Documentation.

2 Scapy Documentation.

3 Flask Web Framework Documentation.